

Skills Summary

Finding a Percent of a Quantity

Use this reference while completing your worksheet or when studying.

Skill 1 — Benchmark percents on a bar model

Model: the whole bar is 100%. Cut it into ten equal parts and each part is 10%; cut it in half and each part is 50%.

10% of a number = the number $\div$ 10. 50% = half. 5% = half of 10%. 1% = the number $\div$ 100.

Example: 10% of 60

Step 1. Ten equal parts of the bar make the whole, so one part is the whole divided by ten — no multiplication needed.

Step 2. $60 \div 10 = 6$, and ten parts of six do rebuild sixty, so the split is right.

Try it: 10% of 90

check: 6

Skill 2 — Multiply by the decimal

Rule: a percent of a quantity is $(\text{percent} \div 100) \times \text{quantity}$. Writing 35% as 0.35 turns “of” into “ $\times$ ”, so 35% of 80 is 0.35×80 . This route works for *any* percent, benchmark or not.

Example: 35% of 80

Step 1. Thirty-five is not a benchmark I can halve or split off a bar, so I convert it to a decimal and multiply.

Step 2. $35\% = 0.35$, and $0.35 \times 80 = 28$; a check against 10% of $80 = 8$ says the answer should sit near three and a half eights.

Try it: 45% of 60

check: 27

Skill 3 — Use the fraction shortcut

Rule: some percents are simple fractions, and the fraction is faster by hand.

- $50\% = \frac{1}{2}$, $25\% = \frac{1}{4}$, $75\% = \frac{3}{4}$
- $20\% = \frac{1}{5}$, $10\% = \frac{1}{10}$, $12.5\% = \frac{1}{8}$

Example: 75% of 48

Step 1. Seventy-five percent is the friendly fraction three quarters, so dividing and then multiplying beats a decimal multiplication.

Step 2. One quarter is $48 \div 4 = 12$, and three quarters is $3 \times 12 = 36$.

Try it: 20% of 85

check: 17

Skill 4 — Percents in a story

How to set it up: find the **whole** the percent is taken *of*, then take the percent of that whole.

A tip or tax is *added* to the whole; a discount is *taken off* it. When two percents follow each other, the second one acts on the new amount, not the original.

Example: An 18% tip on a 50 dollar bill

Step 1. The tip is a percent of the bill, so the bill is the whole and I take eighteen percent of it.

Step 2. $18\% = 0.18$, and $0.18 \times 50 = 9$ dollars of tip.

Try it: A 40 dollar shirt is 15% off. What is the sale price?

check: 34 dollars